Day 14 Evidence Workbooklet

Nested-Loop Tracing at Trace and Debug Level

Engineering practice to row and column traces to inner reset to nested output evidence

Student copy. Guided workbooklet, not the mastery quiz. Print format: duplex, left-stapled packet.

Name		**Section**		**Date**	
Score	______/44		Mastery check	secure / developing / needs support	

Concrete engineering situation

The lab now checks multiple stations and multiple kit slots at each station. Day 14 introduces nested loops at the trace/debug level: an outer loop selects the station and an inner loop checks slots inside that station.

Engineering task	Trace a station-by-slot inspection grid without losing which loop variable is changing.
Computational move	Use an outer loop and inner loop to represent repeated work inside repeated work.
Python focus	nested for loops, outer variable, inner variable, output order, per-row count, reset location, and nested-loop bug evidence.
Evidence to keep	outer value, inner value, execution order, printed output, per-station count, reset before inner loop, and final summary.

Day 14 working rules

1. The outer loop chooses one group.
2. The inner loop completes all of its passes for that group.
3. Then the outer loop moves to the next group.
4. A per-group counter should reset before the inner loop.
5. Trace output in the actual order Python prints it.

Point map **Manage your evidence** blueprint

Points	Section	Evidence focus
1	Read outer and inner loop variables	recognize, predict
2	Trace nested-loop output order	trace, output
3	Use a per-station counter with a reset	state, reset
4	Debug a nested counter bug	debug, state
5	Connect nested loops to a grid of evidence	grid, explain
6	Transfer and support route	transfer, reflect

1 Read outer and inner loop variables

recognize predict

Identify the two loop variables before trying to trace output.

```
for station in range(2):
    for slot in range(3):
        print(station, slot)
```

Pts.	Prompt	My evidence / response
1	What values does station take?	
1	What values does slot take for each station?	
1	How many total print lines appear?	
1	Which loop is the inner loop?	

2 Trace nested-loop output order

trace output

Write the output lines in exact order. The inner loop finishes before station changes.

```
for station in range(2):
    for slot in range(3):
        print("check", station, slot)
```

Pts.	Prompt	My evidence / response
2	Line 1 and line 2 of the output.	
2	Line 3 and line 4 of the output.	
2	Line 5 and line 6 of the output.	
2	What changes fastest: station or slot?	
2	Why does slot return to 0 after station changes?	

3 Use a per-station counter with a reset

state reset

A per-station count should reset once per station, before checking that station's slots.

```
for station in range(2):
    station_count = 0
    for slot in range(3):
        station_count = station_count + 1
    print(station, station_count)
```

Pts.	Prompt	My evidence / response
2	Where is station_count reset?	
2	What value prints for station 0?	
2	What value prints for station 1?	
2	Why would the final values be wrong if station_count were initialized before the outer loop?	
2	Write a reset-location evidence sentence.	

4 Debug a nested counter bug

debug state

The intended result is each station prints a count of 3. This version prints a growing total.

```
station_count = 0
for station in range(2):
    for slot in range(3):
        station_count = station_count + 1
    print(station, station_count)
```

Pts.	Prompt	My evidence / response
2	What count prints for station 0?	
2	What count prints for station 1?	
2	What should station 1 print if the count is per station?	
2	What is the likely cause?	
2	Where should the reset line move?	

Common mistake to avoid

Counters for one group usually reset before the inner loop for that group.

5 Connect nested loops to a grid of evidence

grid explain

A nested loop is a code version of a row-by-column grid. Use the grid idea to explain the loop.

Pts.	Prompt	My evidence / response
2	In a station-by-slot grid, what does the outer loop represent?	
2	What does the inner loop represent?	
2	How many checks happen for 3 stations with 4 slots each?	
2	Which number belongs in range if stations are numbered 0, 1, 2?	
2	Write one sentence explaining why nested loops are harder to trace than one loop.	

6 Transfer and support route

transfer reflect

Close Day 14 by naming the nested-loop evidence you will need before adding lists.

Pts.	Prompt	My evidence / response
4	Write a short note explaining how an outer loop and inner loop work together.	
2	Circle your support route: outer value / inner value / output order / counter reset / total checks. Name one next action.	

Support route

Day 15 adds list state: instead of only using range values, the loop can read stored values from a list by position or by item.